COMPREHENSIVE NUTRITION COURSE

Pregnancy Care

Nutrition & Lactation

PRESENTED BY

Dr. Ahmed El-Khateeb

- Assistant Professor of Pharmacology · Faculty of Medicine
- Consultant of Obesity & Clinical Nutrition
- Certified Life Coach

Agenda

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NORMAL CHANGES IN MATERNAL PHYSIOLOGY

BLOOD VOLUME

↑ 20 %

increases

PLASMA VOLUME

↑ 50 %

increases

EDEMA · OCCURS IN

60–75 %

of women

HEMODILUTION

Concentrations of most vitamins and minerals in blood **decrease**

Blood lipid levels : ↑ in concentrations of

- cholesterol
- LDL
- Triglycerides
- HDL cholesterol

BLOOD GLUCOSE LEVELS

Increased insulin resistance

↑ (increased plasma levels of glucose and insulin)

GASTROINTESTINAL CHANGES

01



Relaxed GIT muscle tone

02



↑ gastric and intestinal transit time

03



Nausea (70%) vomiting (40%)

04



Heartburn

05



Constipation

KIDNEY

Kidney Changes



Increased risk of urinary tract infection

FOOD INTAKE

- increased appetite and food intake; weight gain
- Taste and odor changes, modification in preference for some foods
- Increased thirst

IMMUNE SYSTEM

- Suppressed immunity
- Increased risk of urinary and reproductive tract infection

- Increased cardiac output through ↑ heart rate and stroke volume (30-50%)
- Decreased blood pressure in the first half of pregnancy (9%), followed by a return to non pregnancy levels in the second half

HORMONES

- Placental secretion of Large amounts of hormones needed to support physiological changes of pregnancy

PREPARATION OF HEALTHY PREGNANCY

01



Achieve & maintain healthy BWt

02



Choose balanced Diet

03



be physically Active

04



Regular Health Care Visits

05



Avoid harmful effects

UNDER WEIGHT

Advised To gain weight before pregnancy

Under weight → **Low birth weight** →

- Decrease IQ
- Education Problem
- Short Stature

OVER WEIGHT

- Infant of Obese female is larger > Normal , may Not receive special medical Care
- Obese women → gestational DM .. HTN - infection

NUTRIENT NEEDS DURING PREGNANCY

Each organ of fetus needs nutrients most during its own growth period

Deficiency during one stage may affect heart, during another stage, the developing limbs

ENERGY

ENERGY

C. weight

X 25

FIRST TRIMESTER

no change

SECOND TRIMESTER

350

increases kcal/day

THIRD TRIMESTER

450

increases kcal/day



NUTRIENT NEEDS DURING PREGNANCY

1 — PROTEIN

1.2-1.5

g \ kg

2 — CARBOHYDRATE

45-65%

3 — FAT

25-35%

- essential fatty acids is important for Growth of fetus
- Best is Omega 3,6 for neurological & eye development
- Good Sources : Oily Fish : (Salmon – Mackerel – Sardines)

- Protein unit corrected: the handout reads "1.2–1.5 mg \ kg" — protein is given in g / kg.



04 VITAMINS

VITAMIN B-12

↑10%

needs increase about 10%

FOLATE

400 → 600 µg

DRI increases by half from 400 to 600 micrograms

FOLATE REQUIREMENTS IN PREGNANCY

Adequate folate is critical **before** and **during the first 4 weeks** of pregnancy.



FOLATE REQUIREMENTS

UNPLANNED
50 %

GESTATION
8 wk

Since 50% of pregnancies are unplanned and most women do not seek prenatal care until 8 weeks gestation, **folate supplements prior to conception are critical to prevent NTD.**

600 µg **DRI · pregnancy**

400 µg **for non pregnant woman**

GOOD SOURCES

- Beans
- peas
- orange juice
- green leafy vegetables
- fortified cereals



05 CALCIUM & VITAMIN D

Intestinal absorption of calcium **doubles** during pregnancy

Mother's bone stores are drawn upon as fetal bones begin to calcify

Vit D increase Abs of Ca .. its deficiency → Rickets , Teeth abn.

Vitamin D Fortified milk is sufficient to support normal Ca Metabolism



06 P Mg

Phosphorus & Magnesium

Needed for developing bone

07 Zn

Zinc

Required for DNA and RNA synthesis

Sources : meat & nuts

RDA

11 mg / day

12 mg / day ≤ 18 y

- Zinc RDA in pregnancy: 11 mg/day (19–50 y); the handout's 12 mg is the 14–18 y value · NIH ODS / IOM DRI.



08 FLUIDS

FLUIDS

8-10

Cups \ day

↓ IF FLUID INTAKE ↓ →

- Retention
- UTI
- Constipation
- Dehydration

SOURCES Water > Soups – Juice



09 IRON

⚠ DEFICIENCY INCREASES RISK OF

- maternal and infant death
- preterm delivery
- low birth weight babies

IRON TREATMENT RECOMMENDATIONS · IRON-RICH FOODS

- Meat, fish, poultry, eggs
- Organ meats
- Peas and beans
- Dried fruit
- Whole grain and enriched cereal
- Vitamin C ↑ Abs



10 C Vitamin C

RDA 85 mg — 80 mg ≤ 18 y

- ↓ If decreased → Premature Birth
- Source : Citrus Fruit Or Juice

11 A Vitamin A

RDA 770 µg — 750 µg ≤ 18 y

- ↑ If increased : TERATOGENIC
- ↓ If Decreased : LBW , growth problem , Preterm delivery

- RDAs in pregnancy (19–50 y), NIH ODS / IOM DRI: Vitamin C 85 mg (handout: 80 mg = 14–18 y); Vitamin A 770 µg RAE (handout: "750 mg" — unit is µg; 750 µg = 14–18 y).



EXERCISE DURING PREGNANCY

BENEFITS OF EXERCISE DURING PREGNANCY

- Helps reduce backaches, constipation, bloating, and swelling
- **May help prevent or treat gestational diabetes**
- Increases energy, improves mood and sleep
- Improves posture, promotes muscle tone, strength, and endurance

EXERCISE DURING PREGNANCY

Acceptable activities

- Walking
- dancing
- biking
- Swimming
- Yoga

Exercises to avoid

- Scuba Diving
- Trampoline
- Contact Sports (Basketball)

WARNING SIGNS TO STOP EXERCISE

STOP

01



Vaginal bleeding

02



uterine contractions

03



decreased fetal movement

04



fluid leaking from the vagina

05



Dizziness or feeling faint

06



shortness of breath

07



Chest pain

08



headache

09



muscle weakness

10



calf pain

PRACTICES TO AVOID



02 Cigarette smoking

Restricts blood supply to growing fetus

Increases iron needs

Decreases availability of :

- Vitamin B-12
- Vitamin C
- Folate
- Zinc

01 Do not try to diet during pregnancy

03 Caffeine

Crosses placenta & May retard fetal growth

Heavy consumption may increase risk of spontaneous abortion

04 Drugs :



MEDICAL COMPLICATIONS AND OTHER PROBLEMS OF PREGNANCY

1 GESTATIONAL DIABETES

= appearance of abnormal glucose tolerance during pregnancy with subsequent return to normal postpartum

RISK FACTORS

1. Age 30 or older
2. Obesity
3. Complications in previous pregnancies
4. Family history of diabetes

& Without proper management, can lead to infant sickness and death



GESTATIONAL DIABETES

MEDICAL NUTRITION THERAPY

01

Select foods rich in complex carbohydrates

02

Limit intakes of concentrated sweets

03

Provide for optimal protein intake

04

Three meals daily + snacks

2 OXIDATIVE STRESS AND ENDOTHELIAL DYSFUNCTION

Oxidative stress → damages the endothelium → **Endothelial dysfunction** →

- ↓ blood flow
- ↑ tendency for blood to clot
- increases plaque formation

→ **hypertensive disorders of pregnancy**

TREATMENT supply of antioxidants.

3 PRE-ECLAMPSIA

a condition characterized by

01



High blood pressure

02



Generalized fluid retention

03



Protein in the urine

4 NAUSEA OF MORNING SICKNESS

Arises from normal hormonal changes taking place early in pregnancy

HELPFUL STRATEGIES

- On waking, **arise slowly**
- **Eat dry toast** or crackers
- Eat **small frequent** meals
- If some food odors are offensive, avoid those foods
- When nauseated, drink no citrus juice, milk, coffee or tea
- **Vitamin B6 (pyridoxine) supplements**
- Ginger in doses of 1 gram a day for 4 days decreases nausea and vomiting
- Multivitamin supplements taken prior to and early in pregnancy decreased N & V

- Ginger dose: the handout reads "in doses of gram a day" — the number was missing; 1 g/day for 4 days (Vutyavanich et al., Obstet Gynecol 2001).



CONSTIPATION

Results from Hormones of pregnancy alter muscle tone

TREATMENT

foods high in fiber
(raw fruits and vegetables, whole-grain cereals)

Exercise daily

Drink (at least 8 glasses a day), including fruit juices



HEARTBURN

Is a burning sensation in the lower esophagus, near the heart

Growing infant may put pressure on the woman's stomach, causing acid to back up

TREATMENT

Eat **small frequent** meals

Drink **liquids** between meals

Avoid **spicy** or greasy foods

Sit up while eating

Wait an hour after eating before lying down

Wait two hours after eating before exercise

حلول غير دوائية

COMPLAINT

Pica

Nausea Vomiting

1. أول ما تصحى تاكل ف السرير .. بعدها تفطر بربع ساعة
2. وجبات خفيفة كل ساعتين
3. ↑ CHO , ↓↓ fat
4. Avoid smelling of food
   يعني متطبخش ☺
5. تشرب $\frac{1}{2}$ ساعة قبل & $\frac{1}{2}$ ساعة بعد الوجبة
6. Avoid **C**offee & give **C**old Fluid
7. ادي علاج للـ nausea
   Vitamin B6 25 mg 3 / day
8. وممكن تاخد جنزبيل Ginger على طرف الملعقة

حلول غير دوائية

COMPLAINT

Hyperemesis

في 3 حالات :

- فقدت 5 % وزن
- حصل جفاف
- نقص في الـ K+

1. Parenteral Vitamins (Low in Iron)
2. If multi vitamins → Nausea : Use FA alone
3. IV Fluids
4. Saline + KCl + Thiamine

Constipation

1. Fiber rich Food (Whole Grain - Vegetables - fruits - legumes)
2. 6-8 Cups / day
3. glass of Cold water in Early Morning
4. Eat Dry fruit
5. Exercise : Walking

حلول غير دوائية

COMPLAINT

Heart burn

1. Small , low fat meal & snacks
2. تجنب الأكل الحراق – المقليات – الحمضيات
3. ممنوع المشروبات الغازية
4. avoid antacid ? as it bind to iron
5. Wear loose Clothes

Flatulence

Avoid Cabbage – Garlic – Onion – legumes

Lactose Intolerance

Lactose free product – أرز بلبن – بشاميل – كيك – جبنة

Fe



حلول غير دوائية

COMPLAINT · ANAEMIA

لو ظهرت أعراض نقص الحديد

نزود

- لحمة
- بيض
- خضراوات داكنة
- بقوليات

RDA OF IRON IN PREGNANCY

27 mg (3 mg Abs)

- Iron RDA in pregnancy is 27 mg/day (IOM DRI 2001, NIH ODS); the handout's 30 mg is the older 1989 RDA.



LACTATION

Continue to eat nutrient food , Not restricting weight gain

MILK PRODUCTION NEEDS

700-800

cal \ day

PRODUCING MILK COSTS WOMEN

500 Cal \ Day

above her regular needs during 1st 6th month

1st 6 month + 450 Cal & **2nd 6 month** + 300 cal



LACTATION · REQUIREMENT

PROTEIN

1.5

at least – g \ kg

CARBOHYDRATE

45-65%

OMEGA 3 FA

in fish or supplement

FLUIDS

14 cup\day

- water
- milk
- juice

to protect against dehydration

NB Onion & garlic → alter flavour of breast milk

LACTATION · WEIGHT LOSS

Neither **QUALITY** nor **QUANTITY** of breast milk is adversely affected by **MODERATE** weight loss & infant grows normally .

500 gm weight loss \ week is safe and won't reduce milk out put

physical activity is best way to reduce weight .

Nutritional deprivation only decrease Quantity not quality of milk

Ex : Ca , dietary calcium has no effect on ca Concentration of breast milk.

LACTATION · MICRONUTRIENTS

↑↑ Micronutrients over requirement of pregnancy :

A	B12	C	E	B7
Vitamin A	Vitamin B 12	Vitamin C	Vitamin E	biotin
Cr	I	Zn	Se	B2
chromium	iodine	zinc	selenium	riboflavin

فوائد الرضاعة الطبيعية

BREASTFEEDING

للطفل

- تغذية ومثالية جاهزة دائما
- غذاء معقم وسهل الهضم
- تقوية عضلات الفم والنمو السليم لفك الطفل
- ازدياد الارتباط بالأم
- الوقاية من أمراض الحساسية الغذائية

للأم

- تقوية العلاقة
- تمنع تحجر الثدي
- تساعد على عودة الرحم للحجم الطبيعي بعد الولادة بفضل إفراز هرمون الأوكسيتوسين
- تساعد الأم على سرعة استعادة وزنها الطبيعي
- انخفاض معدل الإصابة بسرطان الثدي والمبيض وهشاشة العظام



أعشاب تزيد من إدرار الحليب

HERBS

1 الشمر

2 ملعقة صغيرة من بذور الشمر بعد طحنها مع كوب ماء مغلي .. وتنقع به البذور لمدة 10 دقائق ويشرب مرتين لـ 3 مرات يوميا

2 الحلبة

1-2 كوب من مغلي بذور الحلبة

3 الكراوية

2-3 كوب يوميا : تطحن بذور الكراوية وتنقع في ماء مغلي لمدة 15 ق

4 شوربة العدس

Thank You

LIFESTYLE FOUNDATION

Dr. Ahmed El-Khateeb
Consultant of Obesity & Clinical Nutrition

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